

Revision of the Grade 7 course

Powers, brackets, factorising and linear equations — the four tools every fraction question in this chapter will need.

LESSON 1–3

3 × 40 MIN

ALGEBRA 8, P. 3

STAGE 8 REVIEW

LESSON CLOCK

5'

12'

6'

15'

2'

Warm-up	5 min
Explanation	12 min
Interactive	6 min
Practice	15 min
Homework	2 min

LEARNING OBJECTIVES

- Apply the laws of indices to simplify products and quotients of powers.
- Expand brackets, including the product of two binomials.
- Factorise using a common factor, grouping, and the three standard identities.
- Solve a linear equation and rearrange a simple formula.

TEXTBOOK

National	pp. 3–6
Cambridge	Stage 9, Unit 2 (pre-work)
Workbook	—

01

Explanation

Why we start here

Everything in Chapter I — cancelling fractions, common denominators, adding and multiplying them — is really *factorising in disguise*. A fraction such as $\frac{x^2 - 9}{x + 3}$ only simplifies once you see $x^2 - 9$ as $(x - 3)(x + 3)$. So three lessons on the Grade 7 toolkit is not lost time; it is the whole quarter's foundation.

1 • The laws of indices

For any $a \neq 0$ and whole numbers m, n :

$$a^m \cdot a^n = a^{m+n} \quad \cdot \quad a^m \div a^n = a^{m-n} \quad \cdot \quad (a^m)^n = a^{mn} \quad \cdot \quad (ab)^n = a^n b^n$$

Multiply powers of the same base — **add** the indices. Divide — **subtract**. A power of a power — **multiply**. Nothing else is allowed: $a^3 + a^3$ is $2a^3$, never a^6 .

THE CLASSIC SLIP

$(2a)^3 = 8a^3$, not $2a^3$ — the index reaches the 2 as well.

2 • Expanding brackets

Every term inside meets every term outside:

$$a(b + c) = ab + ac \quad \cdot \quad (a + b)(c + d) = ac + ad + bc + bd$$

Three products are worth knowing by heart, because you will need to read them *backwards* all quarter:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a - b)(a + b) = a^2 - b^2$$

3 • Factorising — the four moves

1. **Common factor.** $6x^2 - 9x = 3x(2x - 3)$

2. **Difference of two squares.** $4x^2 - 25 = (2x - 5)(2x + 5)$

3. **Perfect square.** $x^2 - 10x + 25 = (x - 5)^2$

4. **Grouping.** $ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$

Always try the common factor first. It is the move learners forget, and it makes the other three easier.

4 • Linear equations and formulae

Clear the fractions, open the brackets, collect the unknowns on one side, divide.

Rearranging a formula is the same process with letters instead of numbers: to make

h the subject of $S = 2\pi rh$, divide both sides by $2\pi r$ to get $h = \frac{S}{2\pi r}$.

02

Worked examples

EXAMPLE 1 Simplify $(3x^2y)^2 \cdot 2xy^3 \div 6x^3y^4$

① $(3x^2y)^2 = 9x^4y^2$

The index 2 reaches the 3, the x^2 and the y .

② $9x^4y^2 \cdot 2xy^3 = 18x^5y^5$

Multiply the numbers, add the indices.

③ $18x^5y^5 \div 6x^3y^4 = 3x^2y$

Divide the numbers, subtract the indices.

ANSWER

$$3x^2y$$

EXAMPLE 2 Factorise $2x^2 + 7x + 3$

① $2 \cdot 3 = 6$

Multiply the first and last coefficients.

② $6 = 1 \cdot 6$, and $1 + 6 = 7$

Find two numbers with product 6 and sum 7.

③ $2x^2 + x + 6x + 3$

Split the middle term using those numbers.

④ $x(2x + 1) + 3(2x + 1)$

Group in pairs and take out the common factor of each pair.

⑤ $(2x + 1)(x + 3)$

The bracket $(2x + 1)$ is now itself a common factor.

ANSWER

$$(2x + 1)(x + 3)$$

EXAMPLE 3 *Solve* $\frac{x-2}{3} = \frac{x+4}{5}$

① $5(x-2) = 3(x+4)$

Cross-multiply — legal because 3 and 5 are never zero.

② $5x - 10 = 3x + 12$

Open both brackets.

③ $2x = 22$

Collect x on the left, numbers on the right.

④ $x = 11$

Divide by 2.

ANSWER

$x = 11$

03 Interactive model

Run through one factorisation together, then hand the mouse to a learner for the next.

Factorising, one step at a time

$$x^2 - 10x + 25$$

① x^2

is $(x)^2$

, 25

and

is 5^2

Look
for
two
perfect
squares.

② Check the middle term:

$2 \cdot x$ ✓

$\cdot 5$

$=$

$10x$

It
matches,
and
the
sign
is
minus.

3 $(x - 5)^2$

So
this
is a
perfect
square,
not
a
difference
of
squares.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Power	Daraja	Степень
Base of a power	Daraja asosi	Основание степени
Index (exponent)	Daraja ko'rsatkichi	Показатель степени
Expand the brackets	Qavslarni ochish	Раскрыть скобки
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Common factor	Umumiy ko'paytuvchi	Общий множитель
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Perfect square	To'la kvadrat	Полный квадрат
Linear equation	Chiziqli tenglama	Линейное уравнение
Subject of a formula	Formuladan izlanayotgan kattalik	Выражаемая величина

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check

Check yourself

 $a^5 \cdot a^3$ equals:

a^8

a^{15}

$2a^8$

a^2

 $(3x)^2$ equals:

$3x^2$

$9x^2$

$6x^2$

$9x$

Which is a difference of two squares?

$x^2 + 9$

$x^2 - 9$

$x^2 - 9x$

$x^2 + 6x + 9$

$x^2 - 6x + 9$ factorises to:

$$(x - 3)(x + 3)$$

$$(x - 9)(x + 1)$$

$$(x - 3)^2$$

$$x(x - 6) + 9$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify: $x^5 \cdot x^3$

ANSWER x^8

2. Simplify: $(a^3)^4$

ANSWER a^{12}

3. Expand: $3(2x - 5)$

ANSWER $6x - 15$

4. Expand: $(x + 4)(x - 4)$

ANSWER $x^2 - 16$

5. Factorise: $6a + 9$

ANSWER $3(2a + 3)$

6. Find the value of $2x^2 - 3x$ when $x = -2$

ANSWER 14

7. Solve: $5x - 7 = 3x + 9$

ANSWER $x = 8$

Medium · the standard question

7 PROBLEMS

1. Simplify: $(2a^2b)^3$

ANSWER $8a^6b^3$

2. Expand and simplify: $(x + 3)^2 - (x - 3)^2$

ANSWER $12x$

3. Factorise: $x^2 - 10x + 25$

ANSWER $(x - 5)^2$

4. Factorise: $4x^2 - 9y^2$

ANSWER $(2x - 3y)(2x + 3y)$

5. Factorise: $ax + ay + bx + by$

ANSWER $(a + b)(x + y)$

6. Solve: $\frac{x - 2}{3} = \frac{x + 4}{5}$

ANSWER $x = 11$

7. Simplify: $a^7 \div a^3 \cdot a^2$

ANSWER a^6

Hard · stretch and reason

7 PROBLEMS

1. *Factorise completely: $x^3 - x$*

ANSWER $x(x - 1)(x + 1)$

2. *Factorise: $2x^2 + 7x + 3$*

ANSWER $(2x + 1)(x + 3)$

3. *Simplify: $(3x^2y)^2 \cdot 2xy^3 \div 6x^3y^4$*

ANSWER $3x^2y$

4. *Factorise completely: $x^4 - 16$*

ANSWER $(x - 2)(x + 2)(x^2 + 4)$

5. *Expand: $(a + b)(a^2 - ab + b^2)$*

ANSWER $a^3 + b^3$

6. *Make x the subject: $a(x - 1) = b(x + 1)$, where $a \neq b$*

ANSWER $x = \frac{a + b}{a - b}$

7. *Factorise: $x^2 - 5x + 6$*

ANSWER $(x - 2)(x - 3)$

07 Homework

Homework — 5 problems

Algebra 8, pp. 3–6 (revision exercises). Written work, about 20 minutes.

1. *Simplify: $(2x^3)^2 \cdot 3x \div 4x^4$ (answer: $3x^3$)*
2. *Expand and simplify: $(2x - 1)(x + 5) - 2x^2$ (answer: $9x - 5$)*
3. *Factorise: $3x^2 - 12$*
4. *Factorise: $x^2 + 8x + 16$*

5. Solve: $\frac{2x + 1}{4} - \frac{x}{3} = 1$